

Driving in the Occupancy World: Vision-Centric 4D Occupancy Forecasting and Planning via World Models for Autonomous Driving

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● **Problem/Objective**

- E2E AD models lack world knowledge, leading to poor generalization and safety in dynamic environments.
- Prior world model research mainly focuses on data generation or pretraining, without direct integration into planning.
- A framework is needed to predict future environments and enable safe, explainable trajectory planning.

● **Contribution/Key Idea**

- Proposed Drive-OccWorld, a vision-centric 4D occupancy forecasting & planning world model.
- Enables continuous forecasting and planning (rollout) for improved robustness.
- Achieves state-of-the-art performance on nuScenes and Lyft benchmarks.

- End-to-end autonomous driving model

$$\mathcal{A}(\{o_{-h}, \dots, o_{-1}, o_0\}, \{\tau_{-h}, \dots, \tau_{-1}, \tau_0\}) = \{\tau_1, \dots, \tau_f\}$$

- Input : Sensor observation $\{o\}$ + trajectory $\{\mathcal{T}\}$
- Output : trajectory $\{\mathcal{T}\}$

- Driving world model

$$\mathcal{W}(\{o_{-h}, \dots, o_{-1}, o_0\}, \{a_{-h}, \dots, a_{-1}, a_0\}) = \{s_1, \dots, s_f\}$$

- Input : Prior observation $\{o\}$ + Ego action $\{a\}$
- Output : Plausible future state $\{s\}$

ViDAR Pre-training



- Drive-OccWorld : World model + Planner

- World model

$$\mathcal{W}(\{o_{-h}, \dots, o_{-1}, o_0\}, \{s_1, \dots, s_{t-1}, s_t\}, \{a_{-h}, \dots, a_{-1}, a_0, \dots, a_{t-1}, a_t\}) = s_{t+1}$$

- Input : Sensor observation $\{o\}$ + 3D occupancy $\{s\}$ + Ego action $\{a\}$
 - Output : Future 3D occupancy $\{s\}$

- Occupancy-based planner

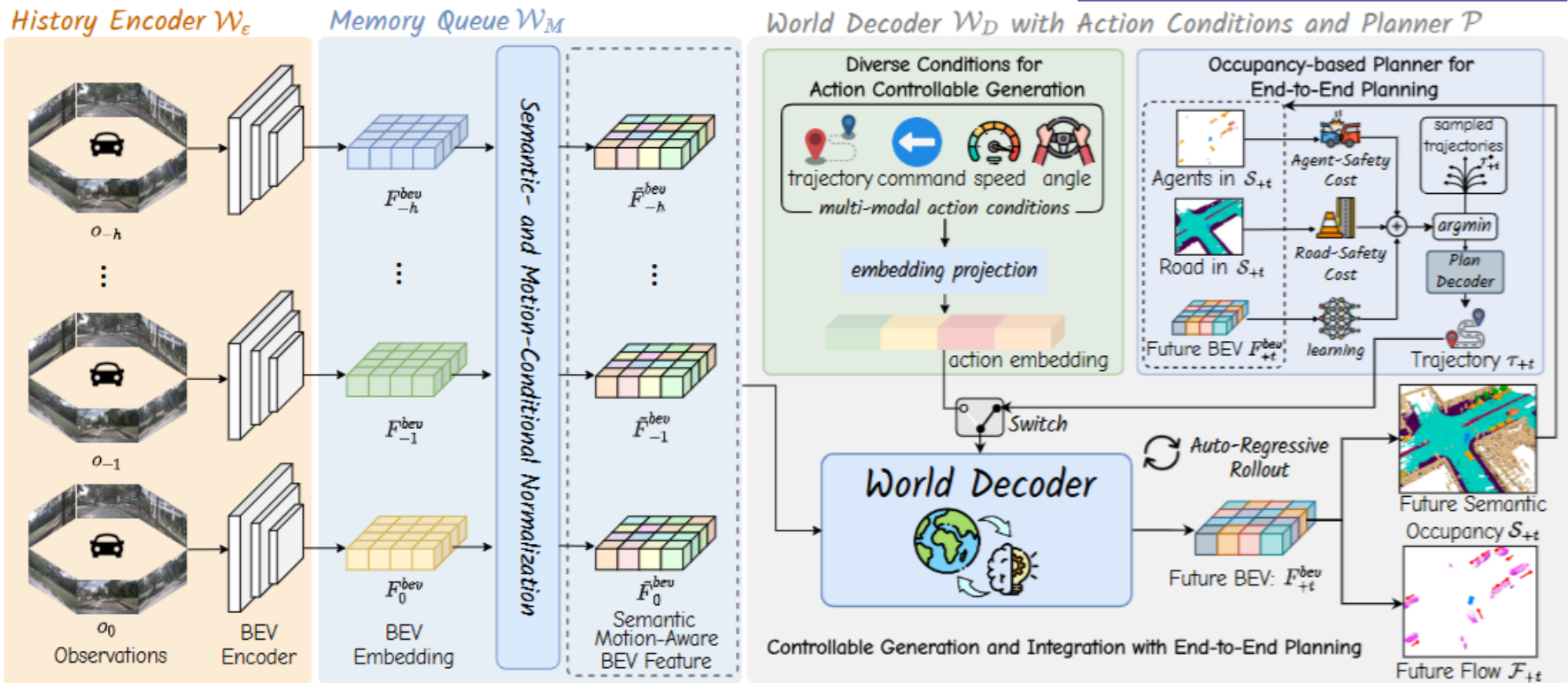
$$\mathcal{P}(f_o(s_{t+1}, \tau_{t+1}^*)) = \tau_{t+1}$$

- Input : 3D occupancy $\{s\}$ + Sampled trajectories $\{\mathcal{T}\}$
 - Output : Predicted trajectories $\{\mathcal{T}\}$

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$$\mathcal{W}(\{o_{-h}, \dots, o_{-1}, o_0\}, \{s_1, \dots, s_{t-1}, s_t\}, \{a_{-h}, \dots, a_{-1}, a_0, \dots, a_{t-1}, a_t\}) = s_{t+1}$$

- Memory Queue : Semantic- and Motion-Conditional Normalization

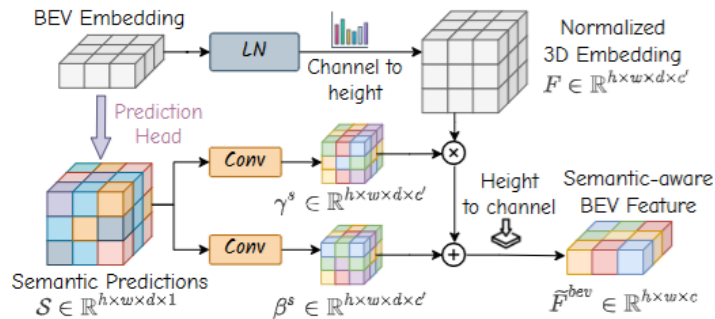
$$\tilde{F}^{bev} = \gamma^* \cdot LayerNorm(F^{bev}) + \beta^*$$

- Semantic-conditional normalization:** BEV 임베딩에 semantic discrimination 강화

1. BEV를 voxel 단위로 분할해, 각 voxel의 **semantic label**을 예측
2. semantic label을 **one-hot embedding**으로 변환
3. Convolution을 통해 (γ_s, β_s) 계산
4. Equation 적용 → semantic-aware BEV 임베딩 생성

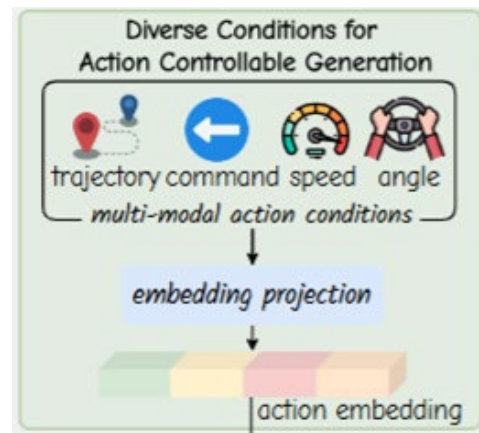
- Motion-conditional normalization**

1. Ego Vehicle Motion - Ego-pose Transformation Matrix
2. Other Agents Motion - 3D backward centripetal flow



- World Decoder : Future forecasting Autoregressive Transformer
 - Inputs :
 - Historical BEV features (from Memory Queue)
 - Expected action condition (velocity, steering, trajectory, command)
 - Process :
 - Learnable BEV queries
 - Deformable self-attention
 - Temporal cross-attention with historical features
 - Conditional cross-attention with action embeddings*
 - Feed-forward network [1]
 - Outputs :
 - Semantic occupancy prediction (voxel-level)
 - 3D backward centripetal flow prediction

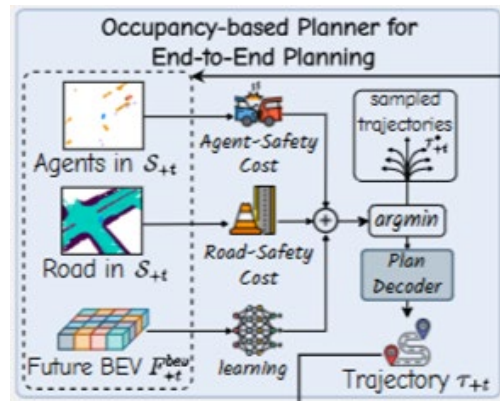
- Action-controllable generation
 - **Motivation:** Ego-vehicle motion states are crucial for modeling interactions with the environment.
 - **Diverse Action Conditions**
 - **Velocity** – ego speed along x, y axes (m/s)
 - **Steering Angle** – converted to curvature (1/m)
 - **Trajectory** – ego displacement (Δx , Δy)
 - **Commands** – go forward / turn left / turn right
 - **Unified Conditioning Interface [2]**
 - Encode actions via Fourier embeddings [3]
 - Concatenate and fuse with learned projections
 - Integrated via **conditional cross-attention** in WD
 - **Key Benefit**
 - Enables **controllable generation**: simulate “what happens if we take this action”



[2] Vista: A Generalizable Driving World Model with High Fidelity and Versatile Controllability

[3] Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains

- End-to-end planning: Occupancy-based planner.
 - **Cost Function Components**
 - **Agent-Safety Cost** – avoid collisions & maintain safe distance from agents
 - **Road-Safety Cost** – penalize trajectories outside drivable areas
 - **Learned-Volume Cost** [4] – learned cost map for complex environments
 - **Planning Process** [4]
 - Sample candidate trajectories guided by high-level commands
 - Select optimal trajectory by minimizing total cost
 - **BEV Refinement**: refine trajectory via cross-attention with BEV features
 - Encode selected trajectory + command → **ego query**
 - Cross-attention with BEV embeddings for fine-grained features
 - Final trajectory predicted via MLPs

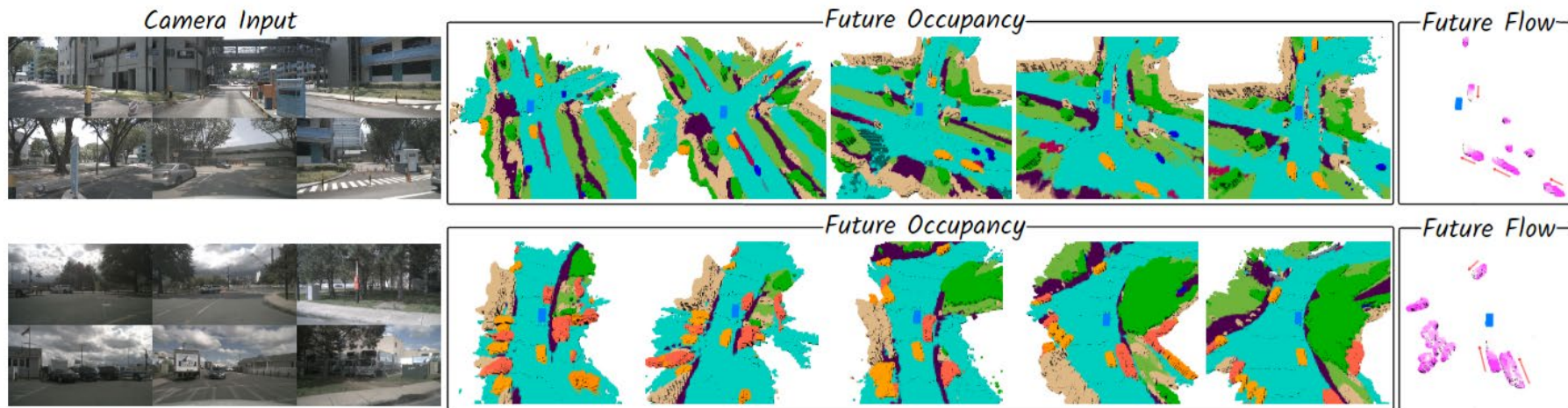


- Experiments: Inflated GMO and Flow Forecasting

Method	Inflated GMO								Fine-Grained GMO		
	nuScenes				Lyft-Level5				nuScenes-Occupancy		
	mIoU _c	mIoU _f (2 s)	mIoU _f	VPQ _f	IoU _c	mIoU _f (0.8 s)	mIoU _f	VPQ _f	mIoU _c	mIoU _f (2 s)	mIoU _f
SPC	1.3	failed	failed	–	1.4	failed	failed	–	5.9	1.1	1.1
OpenOccupancy-C (Wang et al. 2023c)	12.2	11.5	11.7	–	14.0	13.5	13.7	–	10.8	8.0	8.5
PowerBEV-3D (Li et al. 2023a)	23.1	21.3	21.9	20.0	26.2	24.5	25.1	27.4	5.9	5.3	5.5
Cam4DOcc (Ma et al. 2024b)	31.3	26.8	28.0	18.6	36.4	33.6	34.6	28.2	11.5	9.7	10.1
Drive-OccWorld ^A (Ours)	<u>39.7</u>	36.3	<u>37.3</u>	<u>23.7</u>	<u>40.6</u>	<u>39.3</u>	<u>40.0</u>	<u>32.2</u>	13.6	<u>11.9</u>	<u>12.3</u>
Drive-OccWorld ^P (Ours)	39.8	36.3	37.4	25.1	40.9	39.7	40.6	33.4	13.6	12.0	12.4

SPC: SurroundDepth ([Wei et al. 2023a](#)) + PCPNet ([Luo et al. 2023](#)) + Cylinder3D ([Zhu et al. 2021](#))

- Experiments: Qualitative results of 4D occupancy and flow forecasting



• Experiments : End-to-e

Method	L2 (m) ↓				Collision (%) ↓			
	1s	2s	3s	Avg.	1s	2s	3s	Avg.
NMP (Zeng et al. 2019)	-	-	2.31	-	-	-	1.92	-
SA-NMP (Zeng et al. 2019)	-	-	2.05	-	-	-	1.59	-
FF (Hu et al. 2021)	0.55	1.20	2.54	1.43	0.06	0.17	1.07	0.43
EO (Khurana et al. 2022)	0.67	1.36	2.78	1.60	0.04	0.09	0.88	0.33
ST-P3 [†] (Hu et al. 2022)	1.72	3.26	4.86	3.28	0.44	1.08	3.01	1.51
UniAD [†] (Hu et al. 2023b)	0.48	0.96	1.65	1.03	0.05	0.17	0.71	0.31
VAD-Base [†] (Jiang et al. 2023)	0.54	1.15	1.98	1.22	0.10	0.24	0.96	0.43
OccNet [†] (Tong et al. 2023)	1.29	2.13	2.99	2.14	0.21	0.59	1.37	0.72
Drive-OccWorld ^{P†} (Ours)	0.32	0.75	1.49	0.85	0.05	0.17	0.64	0.29
ST-P3 [‡] (Hu et al. 2022)	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71
UniAD [‡] (Hu et al. 2023b)	0.44	0.67	0.96	0.69	0.04	0.08	0.23	0.12
VAD-Base [‡] (Jiang et al. 2023)	0.41	0.70	1.05	0.72	0.07	0.17	0.41	0.22
Drive-WM [‡] (Wang et al. 2024b)	0.43	0.77	1.20	0.80	0.10	0.21	0.48	0.26
Drive-OccWorld ^{P‡} (Ours)	0.25	0.44	0.72	0.47	0.03	0.08	0.22	0.11
UniAD ^{‡*} (Hu et al. 2023b)	0.20	0.42	0.75	0.46	0.02	0.25	0.84	0.37
VAD-Base ^{‡*} (Jiang et al. 2023)	0.17	0.34	0.60	0.37	0.04	0.27	0.67	0.33
BEV-Planner ^{‡*} (Li et al. 2024b)	0.16	0.32	0.57	0.35	0.00	0.29	0.73	0.34
Drive-OccWorld ^{P‡*} (Ours)	0.17	0.31	0.49	0.32	0.02	0.24	0.62	0.29